

Pratyush Padhy

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EDUCATION

University of California, Irvine

Irvine, CA

Bachelor of Science, Computer Science — GPA: 3.86/4.00

Expected Graduation: 2028

• Dean's Honor List: Fall '25, Winter '26, Spring '26 | **Organizations:** Data@UCI, AI@UCI, Cyber@UCI, Hack@UCI

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS)

Pittsburgh, PA

AI/ML Intern — TCS Medical Robotics Center

Jun. 2026 – Present

- Deployed a **ROS2 Jazzy** + **Gazebo Harmonic** physics simulation of the 6-DOF **AgileX Piper** arm from its **URDF**, resolving mesh-rendering failures by configuring **GZ_SIM_RESOURCE_PATH** to load all 6 DAE meshes.
- Engineered a closed-loop ROS2 state pipeline publishing joint commands at **10 Hz** and reading live end-effector pose from the **/tf** transform tree at 1 Hz, providing real-time State/Action/Reward feedback for RL control.
- Trained a **PPO** policy (Stable Baselines3, **150K** environment steps) in a custom **Gymnasium** environment (**6D** action space, **15D** observation space with dense reward shaping) that improved reach success rate from **0/10** to **10/10** episodes, reducing mean final distance from **63 cm** to **3.7 cm** (within 5 cm target threshold).

Ready Tutor

Irvine, CA

Software Developer Intern

Mar. 2026 – Jun. 2026

- Engineered a scalable **MongoDB/SQLite** schema class with **React** useContext integration, replacing **10+** component-level instances and refactoring **5+** database files to centralize and unify storage across application stack.
- Built a Fall Quarter course catalog with real-time retrieval of **1,200+** UCI courses, enabling students to browse and plan schedules; resolved **30+** bugs including a credits calculation issue, contributing to securing **\$5,500** in funding.
- Redesigned the year planning data model with a **JSON**-stringified array schema, replacing a rigid 4-year structure to support fully dynamic multi-year academic scheduling across fall, winter, spring, and summer quarters.

Robotics for All

Palo Alto, CA

Software Intern

Jun. 2023 – Aug. 2024

- Built an AI-powered chatbot for the Robotics for All website to automate volunteer inquiries, and onboarding information, reducing manual response workload and improving overall user engagement across the platform.
- Deployed a comprehensive asynchronous training system with structured readiness assessments for **24+** instructors, standardizing the onboarding pipeline and ensuring consistent teaching quality across all program locations.
- Coached coding classes of **20** students and designed a parent feedback system to continuously improve curriculum quality; interviewed volunteers to assess readiness; awarded the Presidential Service Award (Gold) for **250+** hours.

PROJECTS

FacePulse | Python, TensorFlow, FastAPI, React, OpenCV, Dlib | face-pulse.netlify.app

Jan. 2026 – Jun. 2026

- **Featured on Handshake AI Showcase**, selected among thousands of student projects for national recognition.
- Evolved from a facial landmark emotion detector into a comparative study of **MLP**, **CNN**, and **VGG16** architectures on **FER2013** (**35,000+** images) across **7** emotion classes, achieving **93%** accuracy using **VGG16** transfer learning.
- Built a full-stack web app with a **FastAPI** + **TensorFlow** backend deployed via **Docker** on **Hugging Face Spaces** and a **React/TypeScript** frontend on Netlify with real-time webcam emotion detection and live probability outputs.
- Implemented an analytics dashboard tracking real-time emotion probability charts, per-session timelines, cumulative mood trends, and emotion detection with smooth animated transitions using **Chart.js** and **Framer Motion**.

Argus-AI | Python, YOLOv8, FastAPI, Next.js, TypeScript, Docker | ai-argus.netlify.app

Apr. 2026 – Jun. 2026

- Trained a custom **YOLOv8s** model on **5,200+** dashcam images across **23** object classes, enabling real-time detection of red light running, stop sign breaches, no-entry violations, and pedestrian proximity risks across urban environments.
- Designed a spatial rule engine that infers traffic violations by analyzing proximity relationships between detected bounding boxes, flagging vehicles crossing stop lines, entering no-entry zones, and pedestrian encroachments.
- Deployed a full-stack application with a **FastAPI** + **Python** backend containerized via **Docker** and hosted on **Google Cloud Run**, paired with a **Next.js/TypeScript** frontend on Netlify with annotated violation reports.

iLab Stanford — RNA-seq Research | R, IGV, Bioinformatics, Statistical Analysis

Dec. 2023 – Jul. 2024

- Analyzed human bulk RNA-seq data for non-small cell lung cancer (**GSE268175**), identifying differential expression patterns of **UQCR11** and **OAZ1** across chromosomes and mapping genome-wide expression profiles using **IGV**.
- Generated dimensionality reduction heatmaps and statistical expression plots in **R** to identify gene clusters and expression outliers in single-cell RNA sequencing data; presented findings to Dr. Qian Wang of Stanford's iLab.

TECHNICAL SKILLS

Languages: Python, Java, R, C++, SQL, HTML, CSS, JavaScript, TypeScript

Libraries & Frameworks: TensorFlow, PyTorch, OpenCV, NumPy, Pandas, Matplotlib, scikit-learn, YOLOv8, Stable Baselines3, Gymnasium, Chart.js, Framer Motion, React, Next.js, FastAPI, Tailwind CSS

Tools & Platforms: Git, Docker, ROS2, Gazebo, Google Cloud Run, Hugging Face Spaces, MongoDB, SQLite

Certifications: PCEP – Python, Coursera UCSC – C++, WorldTutor – Java